MANAGEMENT INFORMATION SYSTEMS
ICT2162
LECTURE 08

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Enhancing Decision Making

Intended Learning Objectives

- Identify organizations, Management, and the networked Enterprise.
- Explore Information Technology infrastructure and security of an organization.
- Explain Key System applications for the digital age.
- Identify how to build and manage information systems.

What We Discuss Today?

- Different types of decisions and how does the decision-making process work?
- How do information systems support the activities of managers and management decision making?
- How do business intelligence and business analytics support decision-making?
- How do different decision-making constituencies in an organization use business intelligence?
- What is the role of information systems in helping people working in a group make decisions more efficiently?

Decision Making

Which product is more profitable to the supplier

- Option 01:

  - Sales price of the Casio wristwatch 10,000/=

- Option 02:

  - Sales price of the Citizen wristwatch 18,000/=

Decision Making

Which product is more profitable to the supplier

- Option 01:
  - Sales price of the Casio wristwatch 10,000/=
  - Cost of the Casio wristwatch 6,000/=
- Option 02:
  - Sales price of the Citizen wristwatch 18,000/=
  - Cost of the Citizen wristwatch 10,000/=

Decision Making

Which product is more profitable to the supplier

- Option 01:
  - Sales price of the Casio wristwatch 10,000/=
  - Cost of the Casio wristwatch 6,000/=
  - Sales forecast of the Casio wristwatch 2,000
- Option 02:
  - Sales price of the Citizen wristwatch 18,000/=
  - Cost of the Citizen wristwatch 10,000/=
  - Sales forecast of the Citizen wristwatch 800

Business Value of Improved Decision Making

Improving hundreds of thousands of “small” decisions adds up to large annual value for the business

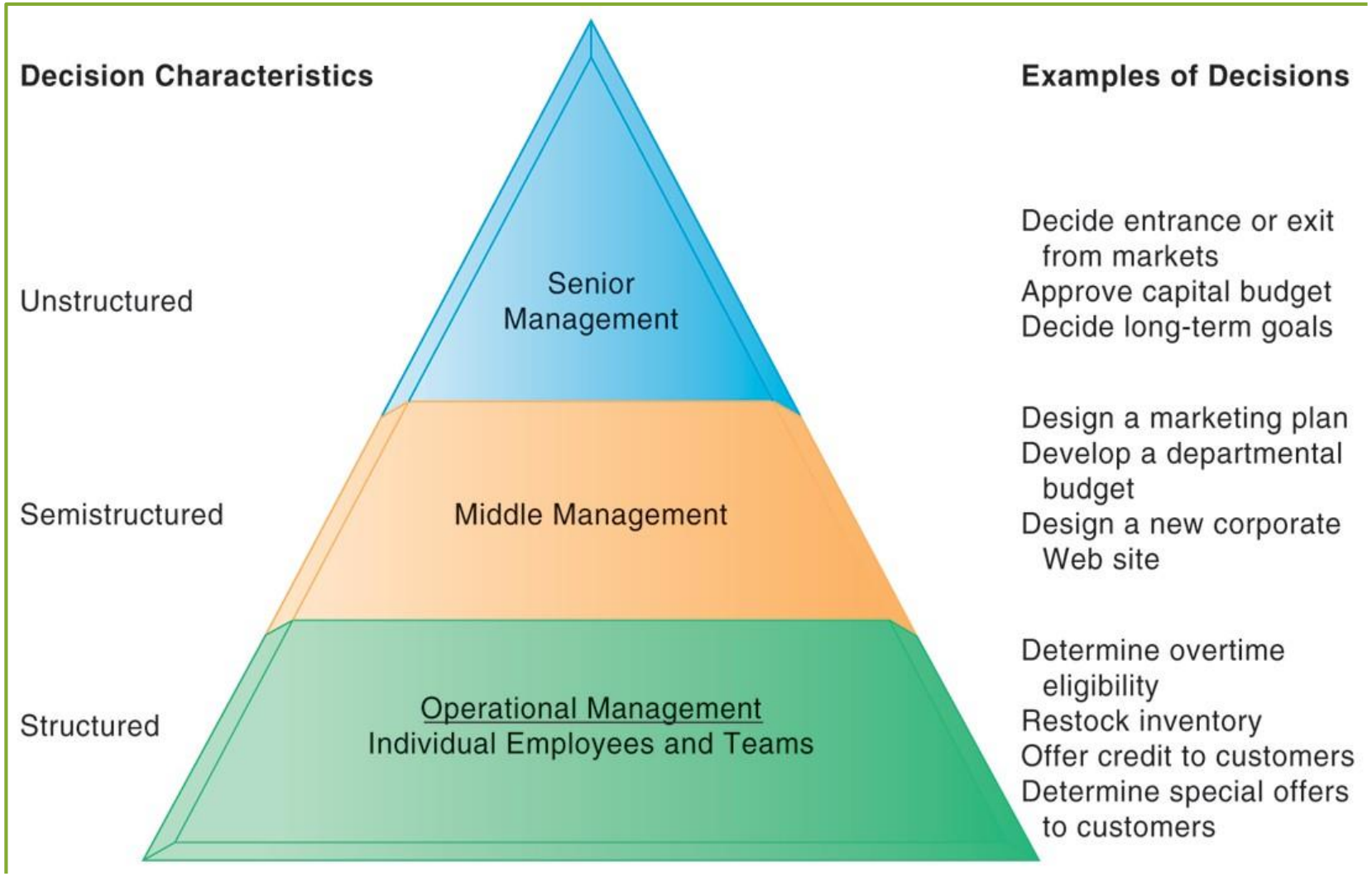
Types of decisions:

1. **Unstructured**:
   - Decision maker must provide judgment, evaluation, and insight to solve problem
2. **Structured**:
   - Repetitive and routine; involve definite procedure for handling so they do not have to be treated each time as new
3. **Semi structured**:
   - Only part of problem has clear-cut answer provided by accepted procedure

Key Decision-Making Groups in a Firm

1. **Senior managers:**
   - Make many unstructured decisions
   - E.g. Should we enter to a new market?
2. **Middle managers**:
   - Make more structured decisions but these may include unstructured components
   - E.g. Why is order fulfillment report showing decline in Colombo?
3. **Operational managers:**
   - Make more structured decisions
   - E.g. Does customer meet criteria for credit?

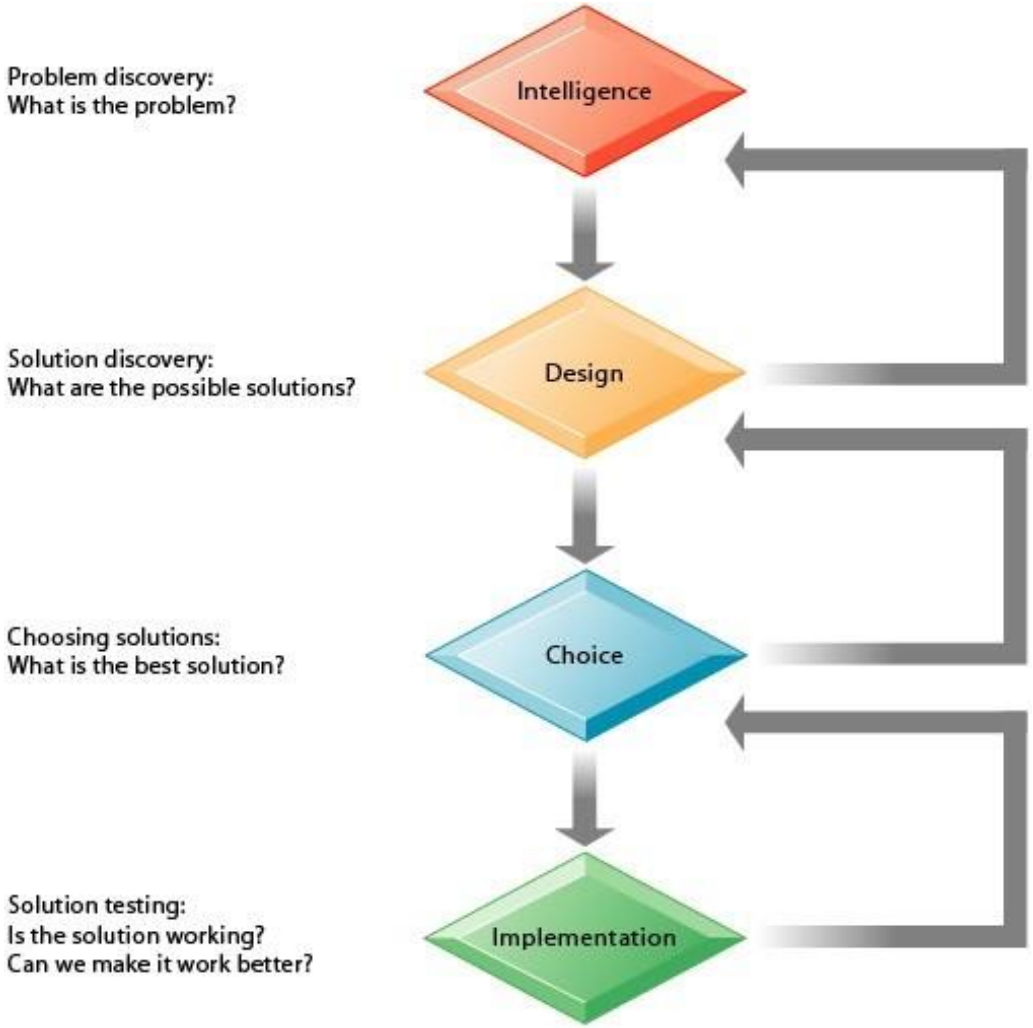
Information Requirements of Key Decision-Making Groups in a Firm



The 4 Stages of the Decision-Making Process

1. **Intelligence**
   - Discovering, identifying, and understanding the problems occurring in the organization
2. **Design**
   - Identifying and exploring solutions to the problem
3. **Choice**
   - Choosing among solution alternatives
4. **Implementation**
   - Making chosen alternative work and continuing to monitor how well solution is working

The 4 Stages of the Decision-Making Process



Managers and Decision Making in the Real World

- Information systems can only assist in some of the roles played by managers, IS cannot improve all the different kinds of decisions taking place in an organization
- **Classical model of management:**

  - 5 functions
    - Planning, organizing, coordinating, deciding, and controlling
- **More contemporary behavioral models:**

  - Actual behavior of managers appears to be less systematic, more informal, less reflective, more reactive, and less well organized than in classical model



Mintzberg's 10 Managerial Roles

- **Interpersonal roles**
  1. Figurehead
  2. Leader
  3. Liaison
- **Informational roles**
  4. Monitor
  5. Disseminator
  6. Spokesperson
- **Decisional roles**
  7. Entrepreneur
  8. Disturbance handler
  9. Resource allocator
  10. Negotiator

Group Activity

Make into three groups and search on internet on below mentioned roles with examples. Present your findings to the class.

- **Interpersonal roles**
  1. Figurehead
  2. Leader
  3. Liaison
- **Informational roles**
  4. Monitor
  5. Disseminator
  6. Spokesperson.
- **Decisional roles**
  7. Entrepreneur
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Real World Decision Making

Three main reasons why investments in information technology do not always produce positive results

01. Information quality

- High-quality decisions require high-quality information

02. Management filters

- Managers have selective attention and have variety of biases that reject information that does not conform to prior conceptions

03. Organizational inertia and politics

- Strong forces within organizations resist making decisions calling for major change

High Velocity Automated Decision Making

- Made possible through computer algorithms precisely defining steps for a highly structured decision
- Humans taken out of decision
  - E.g. High-speed computer trading programs
    - Trades executed in 30 milliseconds
    - Responsible for “Flash Crash” of 2010
- Require safeguards to ensure proper operation and regulation

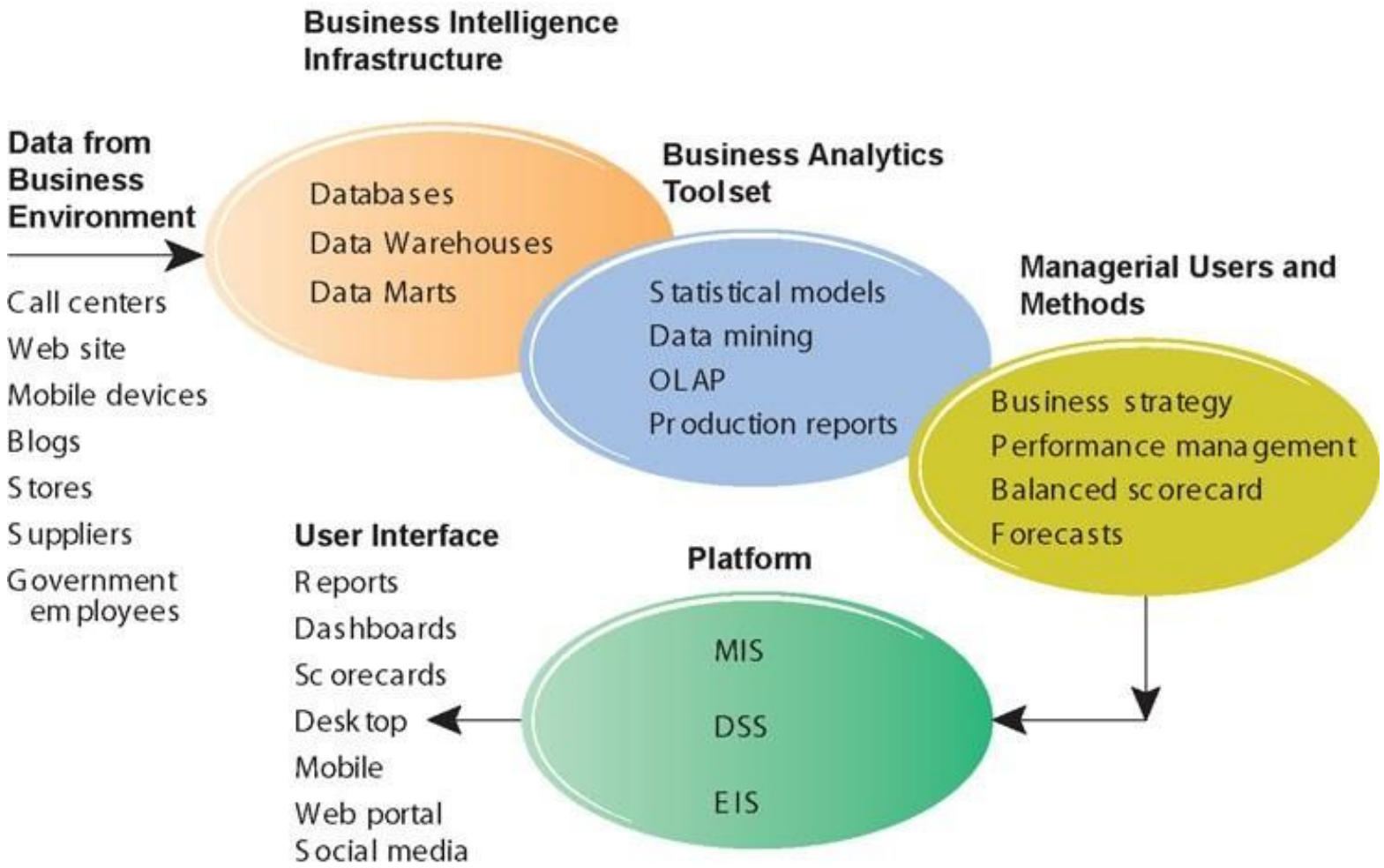
Business Intelligence, Business Analytics and Business Intelligence Vendors

- **Business intelligence**
  - Infrastructure for collecting, storing, analyzing data produced by business
  - Databases, data warehouses, data marts
- **Business analytics**
  - Tools and techniques for analyzing data
  - OLAP, statistics, models, data mining
- **Business intelligence vendors**
  - Create business intelligence and analytics purchased by firms

Six Elements in the Business Intelligence Environment

1. Data from the business environment
2. Business intelligence infrastructure
3. Business analytics toolset
4. Managerial users and methods
5. Delivery platform – MIS, DSS, ESS
6. User interface

Business Intelligence and Analytics for Decision Support



Business Intelligence and Analytics Capabilities

- Goal is to deliver accurate real-time information to decision-makers
- Main functionalities of BI systems
    1. Production reports
    2. Parameterized reports
    3. Dashboards/scorecards
    4. Ad hoc query/search/report creation
    5. Drill down
    6. Forecasts, scenarios, models

Who Uses Business Intelligence?

- 80% are casual users relying on production reports
- Senior executives
  - Use monitoring functionalities
- Middle managers and analysts
  - Ad-hoc analysis
- Operational employees
  - Prepackaged reports
  - E.g. sales forecasts, customer satisfaction, loyalty and attrition, supply chain backlog, employee productivity

Examples of BI Applications

- **Predictive Analytics**
  - Use patterns in data to predict future behavior
  - E.g. Credit card companies use predictive analytics to determine customers at risk for leaving
- **Big Data Analytics**
- **Data Visualization**
  - Help users see patterns and relationships that would be difficult to see in text lists
- **Operational Intelligence and Analytics**
- **Location Analytics**
  - To gain business insight from the location component of the data
- **Geographic Information Systems (GIS)**
  - Ties location-related data to maps

References

- Management Information Systems 17th Edition by Laudon, K. C., & Laudon, J. P

Questions ???



Thank you…